



Factors affecting electronic systems

deposited on the substrate surface.

Etching. During the reaction, graphite is etched away, leaving only diamond, as hydrogen's etching rate for graphite is much higher than that for diamond.

Growth control. The plasma shape and the concentration of active species in the chamber are adjusted by varying deposition pressures, affecting the deposition rate. The substrate temperature, typically between 800 and 1200°C, also influences the form, quality, and growth rate of the film. It is difficult to produce high-quality diamond below 750°C due to slow growth rates and many defects. However, above 1200°C, there is a risk of the diamond graphitising.

The MPCVD technique avoids contamination by hot wires, ensuring high-purity diamonds. It offers stable temperature control for consistent reaction conditions. The versatility of MPCVD allows for the introduction of multiple gases into the system, generating stable plasma over large areas and enhancing diamond growth efficiency. It achieves high growth rates with reproducible sample quality and is cost-effective by eliminating the need for hot wires and utilising multiple gases.

Through the innovative techniques of diamond doping and epitaxy, the semiconductor industry hopes to transform the market for semiconductors and power components, particularly for the

automotive, renewable energy, and quantum computing sectors. Positioned at the forefront of the technological advancement, this technique aims to advance semiconductor diamonds to the necessary level of industrial readiness. It will be a key player in our nation's semiconductor industry.

How diamond addresses the limitations of silicon in semiconductors

Despite silicon's prominence in the semiconductor industry, it has several limitations, especially as technology advances and demands increase. These limitations include:

High thermal conductivity. Due to silicon's limited thermal conductivity, it is susceptible to overheating in high-power and high-frequency applications. Research has indicated that most failure processes in packaged chips display temperature-dependent behaviour. The lifetime of the device is reduced by two times for every 10°C increase in junction temperature. Temperature is the leading cause of nearly 50% of malfunctions in modern electronic systems, as shown in the pie chart.

Diamond is being explored as a solution for thermal management in semiconductor packaging due to its superior thermal conductivity and electrical insulation. Its thermal conductivity— $2 \times 10^3 \text{ Wm}^{-1}\text{K}^{-1}$ over a wide temperature range—is five times higher than that of copper and 15 times higher than silicon at room temperature, making it suitable for mobile and aerospace applications. Diamond's low density and high Young's modulus (1000GPa to 1100GPa) increase package reliability. Single-crystal diamonds achieve a thermal conductivity of up to $22 \text{ W/cm} \cdot \text{K}$, while polycrystalline diamonds reach up to $20 \text{ W/cm} \cdot \text{K}$. This makes them ideal for applications that demand precise thermal

dissipation. Additionally, diamond combines electrical insulation with exceptionally high thermal conductivity, allowing for optimal thermal management of semiconductor packaging.

Wide bandgap. According to research, wide bandgap materials are projected to greatly enhance power electronics, despite silicon having been a crucial semiconductor material for more than 50 years. Diamond, with a bandgap of 5.47eV, significantly larger than that of other semiconductors, is able to withstand high temperatures and voltages, making it suitable for use in high-power electronics and radiation detection applications. Diamond substrates, due to their larger bandgap compared to silicon-based semiconductors, allow for a reduction in die size without compromising blocking voltage. This size reduction facilitates the creation of more compact and efficient power electronics devices. Smaller dies result in reduced output capacitance, transistor gate charge, and parasitic capacitances. Compared to traditional silicon semiconductors, this leads directly to faster transition speeds with lower transition losses, less dissipation from capacitors, and reduced gate charge losses at a given level.

Electric breakdown. Silicon's low electric breakdown at relatively low fields and 1.1eV bandgap limits its operating temperature to below 175°C. Diamond, as a semiconductor with a wide bandgap of 5.47eV, proves to be ideal for high-power electronics and radiation detection due to its high temperature and voltage resistance and superior electric breakdown field ($2 \times 10^7 \text{ Vcm}^{-1}$)—much higher than silicon's ($3 \times 10^5 \text{ Vcm}^{-1}$). Moreover, diamond features a saturation drift rate of $1.05 \times 10^7 \text{ cm/s}$ for holes and $1.5 \times 10^7 \text{ cm/s}$ for electrons without